$$y=\frac{\sin(3t)}{1+t^2}$$

$$\hat{y}=\frac{2t.\sin(3t)+\cos(3t).3(1+t^2)}{(1+t^2)^2}$$

$$\hat{y}=\frac{2t.\sin(3t)+3\cos(3t)(1+t^2)}{(1+t^2)(1+t^2)}$$